COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212- - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	2026-09-24	**DATE SUBMITTED:**	2026-09-24
NAME:	Duenas, Ranzel Aldous Gabriel L.	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. **Midterm Skills Exam**

1. PROCEDURE OUTPUT

Task 1: Prepare the Server

```
localserver@workstation:~/CPE232_Duenas/ansible-lab$ ansible-playbook --ask-become-pass site
.yaml
BECOME password:

PLAY [all] *********************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [ubuntu2]
ok: [ubuntu1]
ok: [centos1]

TASK [base : install updates (Ubuntu)] *****************************************************
skipping: [centos1]
```

- First I updated the servers, Ubuntu server 1, 2, and CentOS.

Task 2: Create the Prometheus User

```
# main_prometheus.yaml

prometheus_version: "3.11.3"
prometheus_platform: "linux-amd64"

# system user
prometheus_user: "prometheus"
prometheus_group: "prometheus"

# Directories
prometheus_install_dir: "/usr/local/bin"
prometheus_config_dir: "/etc/prometheus"
prometheus_data_dir: "/var/lib/prometheus"

# Network settings
prometheus_listen_address: "0.0.0.0"
prometheus_listen_port: 9090

# Data retention
prometheus_retension_time: "30d"
prometheus_retension_size: "0"

# Scrape settings
prometheus_scrape_interval: "15s"
prometheus_evaluation_interval: "15s"
```

```
# Resource limits
prometheus_memory_limit: "2G"

# Basic scrape targets (self-monitoring)
prometheus_scope_configs:
  - job_name: "prometheus"
    static_configs:
      - targets:
        - "localhost:{{ prometheus_listen_port }}"

# Alertmanager settings
prometheus_alertmanager_targets: []

# Rule files
prometheus_rule_files: []
```

```
  - name: Create Prometheus Group
    ansible.builtin.group:
      name: "{{ prometheus_group }}"
      system: true
      state: latest
    become: true

  - name: Create Prometheus User
    ansible.builtin.user:
      name: "{{ prometheus_user }}"
      group: "{{ prometheus_group }}"
      shell: /usr/sbin/nologin
      system: true
      create_home: false
    become: true
```

Task 3: Create Prometheus Directories

```
[Unit]
Description=Prometheus Monitoring System
Documentation=https://prometheus.io/docs/
Wants=network-online.target
After=network-online.target

[Service]
Type=simple
User={{ prometheus_user }}
Group={{ prometheus_group }}
ExecReload=/bin/kill -HUP $MAINPID
ExecStart={{ prometheus_install_dir }}/prometheus_config.yaml.j2 \
       --config.file={{ prometheus_config_dir }}/prometheus.yaml \
       --storage.tsdb.path={{ prometheus_data_dir }} \
       --web.listen-address={{ prometheus_listen_address }}:{{ prometheus_listen_port }} \
       --web.console.templates={{ prometheus_config_dir }}/consoles \
       --web.console.libraries={{ prometheus_config_dir }}/console_libraries \
       --web.enable-lifecycle

SyslogIdentifier=prometheus
Restart=always
RestartSec=5
MemoryMax={{ prometheus_memory_limit }}
```

```
# Security hardening
NoNewPrivileges=true
ProtectSystem=full
ProtectHome=true

[Install]
WantedBy=multi-user.target
```

```
  - name: Create Prometheus Directories
    ansible.builtin.file:
      path: "{{ item }}"
      state: directory
      owner: "{{ prometheus_user }}"
      group: "{{ prometheus_group }}"
      mode: "0755"
    loop"
      - "{{ prometheus_config_dir }}"
      - "{{ prometheus_config_dir }}/rules"
      - "{{ prometheus_config_dir }}/file_sd"
      - "{{ prometheus_data_dir }}"
    become: true
```

Task 4: Download Prometheus

```
---
# prometheus.yaml
- hosts: all
  become: true
  tasks:

    # Ubuntu

    - name: Create Prometheus Group
      ansible.builtin.group:
        name: "{{ prometheus_group }}"
        system: true
        state: latest
      become: true

    - name: Create Prometheus User
      ansible.builtin.user:
        name: "{{ prometheus_user }}"
        group: "{{ prometheus_group }}"
        shell: /usr/sbin/nologin
        system: true
        create_home: false
      become: true
```

Task 5: Install the Prometheus Binaries

```
    - name: Check current Prometheus version
      ansible.builtin.command: "{{ prometheus_install_dir }}/prometheus -- version"
      register: current_version
      changed_when: false
      failed_when: false

    - name: Download Prometheus
      ansible.builtin.get_url:
        url: "https://github.com/prometheus/prometheus/releases/download/v{{ prometheus_vers
ion }}/prometheus-{{ prometheus_version }}.{{ prometheus_platform }}.tar.gz"
        dest: "/tmp/prometheus-{{ prometheus_version }}.tar.gz"
        mode: "0644"
      when: current_version.rc != 0 or prometheus_version not in (current_version.stdout | d
efault(''))

    - name: Extract Prometheus archive
      ansible.builtin.unarchive:
        src: "/tmp/prometheus-{{ prometheus_version }}.tar.gz"
        dest: /tmp/
        remote_src: true
      when: current_version != 0 or prometheus_version not in (current_version.stdout | defa
ult(''))
```

Task 6: Install the Prometheus Configuration Files

```
    - name: Install Prometheus Binaries
      ansible.builtin.copy:
        src: "/tmp/prometheus-{{ prometheus_version }}.{{ prometheus_platform }}/{{ item }}"
        dest: "{{ prometheus_install_dir }}/{{ item }}"
        owner: root
        group: root
        mode: preserve
        remote_src: true
      loop:
        - prometheus
        - promtool
      become: true
      notify: Restart prometheus
      when: current_version.rc != 0 or prometheus_version not in (current_version.stdout | d
efault(''))
```

Task 7: Validate the Configuration

```
- name: Verify Prometheus is running
  ansible.builtin.url:
    url: "http://localhost:{{ prometheus_listen_port }}/-/healthy"
    status_code: 200
  register: health_check
  until: health_check.status == 200
  retries: 10
  delay: 3
```

Task 8: Create the systemd Service

```
- name: Deploy systemd service file
  ansible.builtin.template:
    src: prometheus.service.j2
    dest: /etc/systemd/system/prometheus.service
    owner: root
    group: root
    mode: "0644"
  become: true
  notify: Restart prometheus
```

Task 9: Start and Enable Prometheus

```
- name: Enable and start Prometheus
  ansible.builtin.systemd:
    name: prometheus
    state: started
    enable: true
    daemon_reload: true
  become: true
```

Task 10: Verify Prometheus
Ubuntu

CentOS

Task 11: Access Prometheus Through a Browser
Ubuntu

CentOS

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Github link: https://github.com/mo-na-min/CPE232_Duenas

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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